Design layer

the whole layer, as one document

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ROOT DESIGN

Relay: typed MCP exchanges

1 Relay: typed MCP exchanges

1.1 Foundation

Relay carries MCP meaning and owns protocol execution lifetimes.

GOAL

Serve and call native application operations

Expose typed tools, resources, prompts, and completion through MCP server and client operations over stdio and Streamable HTTP. Reuse each tool's definition across declaration, binding, and typed calling.

GOAL

Retain the complete protocol capability scope

Support explicit revision strategies, both MCP authorization roles, and independently verifiable extension boundaries. The modern implementation and the retained compatibility obligations are distinguished in the pending ledger.

NO-GOAL

Own business execution or identity providers

Application effects, retry policy, durable runs, workflow settlement, web login, and general authorization-server operation belong to callers or independently useful libraries. Relay supplies protocol ports.

INVARIANT

Wire ids never own execution

Exchange and invocation identities select authoritative work and response destinations. Repeated JSON-RPC ids cannot merge concurrent work.

`mechanism`

INVARIANT

Admission precedes handler work

Protocol validation, endpoint bearer admission when configured, tool visibility and execution permission, and native argument decoding precede the handler's effect. A rejection produces no tool invocation.

`mechanism`

INVARIANT

Terminal exchanges suppress subsequent output

Completion and closure consume an exchange in one authoritative reducer history. A late completion or progress report cannot answer a replacement request.

`mechanism`

PRINCIPLE

Keep authority with the boundary that can prove it

Revision codecs own wire meaning; the runtime owns serialized effect interpretation; a token verifier owns token validity; the application owns business policy and effect outcome.

PRINCIPLE

Retain distinct absence and failure meanings

Omitted fields, explicit null, false hints, tool refusals, input requests, protocol failures, and uncertain delivery retain distinct representations. See `ADR-0002` and `ADR-0005`.

1.2 Pending updates

ruling **Bind paging and input state to the admitted view** 2026-10-06

The implemented cursor authenticates family and offset with the server key; continuation state authenticates the operation family and a nonce, without binding arguments, principal or outstanding response keys. The retained design requires catalog revision, tenant and authorized visible-view binding. Cross-trust-domain grant reuse, anonymous access, and policy changes between rounds need explicit semantics before claiming these stronger guarantees.

ADR-0004

ruling **Broaden dynamic schema interoperability deliberately** 2026-10-06

MCP declarations may contain valid JSON Schema outside Blueprint’s finite input-contract profile. UnsupportedSchema remains explicit; valid default-open schemas outside that profile are refused by Blueprint admission. Choosing a broader bounded contract port or an enlarged profile requires separate approval and external peer evidence.

ADR-0002

build **Implement retained revision and OAuth strategies** 2026-10-06

Legacy 2025-11-25 sessions and their transport strategy, older retained revisions, client OAuth discovery/registration/PKCE/refresh flows, and selected extensions remain designed scope. Modern readiness never invents a legacy handshake.

ADR-0001

build **Complete protocol and public test support** 2026-10-06

Resource-read continuation, an approved typed logging emitter, expanded paired/scripted/fake-clock transport and authorization-server helpers, and independently pinned client/auth/interoperability acceptance remain incomplete. Tasks, Apps, sender constraints, workload identity, target expansion, and release hardening require their scoped requirements and evidence.

ADR-0007

verify **Verify resource claims at their actual boundaries** 2026-10-06

Request slots are acquired after Mist reads the bounded body; mounted callers pre-read it themselves. These limits do not establish a bound on incomplete body readers or all network processes. Comprehensive owner-death/process-count evidence, SSE reconnection/resumption where supported, and continuous drift/soak/fuzz coverage remain separate obligations.

ADR-0003

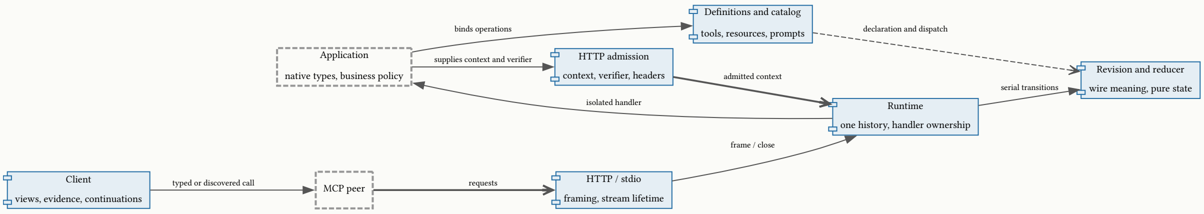
verify **Check carried-over configuration claims against the code** 2026-10-17

The defaults, setters and module map in Configuration surface, and the profile rules in Validation and extension ownership, were moved from retired guides. Only some defaults were compared with the source. Each value and setter name is a claim to confirm against the code, section by section.

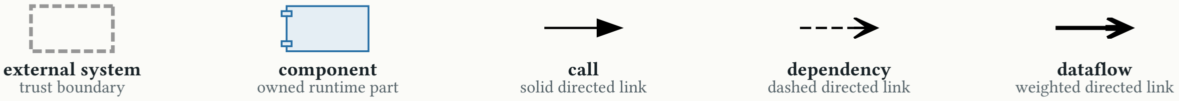
1.3 System at a glance

One MCP context has separate wire, admission, execution, and application owners.

VIEWPOINT CONTEXT-OWNERSHIP · ALTITUDE L2 · CONTEXTS From native contracts to owned exchanges


Definitions and catalog
tools, resources, prompts
binds operations
declaration and dispatch
Application
native types, business policy
supplies context and verifier
HTTP admission
context, verifier, headers
admitted context
Revision and reducer
wire meaning, pure state
serial transitions
isolated handler
Runtime
one history, handler ownership
frame / close
Client
views, evidence, continuations
typed or discovered call
MCP peer
requests
HTTP / stdio
framing, stream lifetime

LEGEND


external system
trust boundary
component
owned runtime part
call
solid directed link
dependency
dashed directed link
dataflow
weighted directed link

Blueprint owns application codecs/contracts, HTTP Gun owns outbound HTTP, and Sinal owns observation delivery. Relay has no dependency on Fabric, Saga, or Hellhound.

- The **Server description** is immutable. An **Endpoint** or **Runtime** owns the live registry, while a copied description has no authority to mutate running work.
- Modern HTTP creates a runtime per request from the endpoint's current server snapshot. Stdio and in-process clients retain one connection runtime. The **Reducer** consumes events in that owner's serialized history.
- Relay is one Erlang distribution with internal protocol, schema, transport, and foreign-boundary modules. A pure transition boundary is retained; JavaScript runtime support is not implemented. See **ADR-0001**.

Decision	Authority	Outcome and boundary
Wire revision	Frozen MCP schema and requirements	Admission never infers compatibility from a malformed response.
Tool input/output	Definition codecs	Erasure retains closures; it does not expose Dynamic to ordinary callers.
Bearer validity	Configured verifier	Attestation asserted by trusted adapter; Relay checks resource/scopes.
Execution history	Runtime actor	One interpreter starts each admitted invocation; copied pure values do not prove exactly-once effects.
Business retry and durability	Application	Submission evidence and cancellation never establish rollback.

Public module	Purpose
`relay/tool`	Tool definitions, handlers, the `Call` a handler sees, and declarations.
`relay/resources`, `relay/prompts`, `relay/completion`	The other server services.
`relay/content`, `relay/subscriptions`	Protocol data.
`relay/server`	The server description.
`relay/http`, `relay/stdio`	The server transports.
`relay/client`, `relay/client/output`	The client and its optional typed-output projection.
`relay/authorization`	Bearer tokens, admission, challenges and metadata.
`relay/telemetry`	Sinal event descriptors.
`relay/runtime`, `relay/reducer`	Custom transports: the runtime actor and the pure reducer.
`relay/testing`	An in-process client, MCP requests for a mounted handler, and test verifiers.

1.4 Domain model

Exchange ownership, handler execution, and connection lifetime are independent state axes.

- A Definition may create several Bound tool values. Each Exchange keeps a fresh response destination; each Invocation keeps its own handler identity even when a Request id repeats.

Definition

value-object · immutable · MCP · owned by tool admission

A tool's native contract reused on both sides of MCP.

ATTRIBUTES

authored **Contract** · name × Codec(input) × output mode
Structured output retains a codec and schema; content-only output retains no output codec.

derived **Declaration** · schemas, annotations, icons, metadata
Derived from admitted codecs and the caller's independent optional settings.

RELATIONSHIPS

`1 : 0..n` One definition may bind several handlers or serve typed calls without changing native type agreement.

Exchange phase

Vacant · Active · Stream · Done

Exchange

entity · stateful · MCP · owned by reducer history

An inbound frame's authoritative destination, independent of its wire id.

ATTRIBUTES

derived **Identity** · opaque ExchangeId
The transport allocates a fresh node-unique identity before parsing.

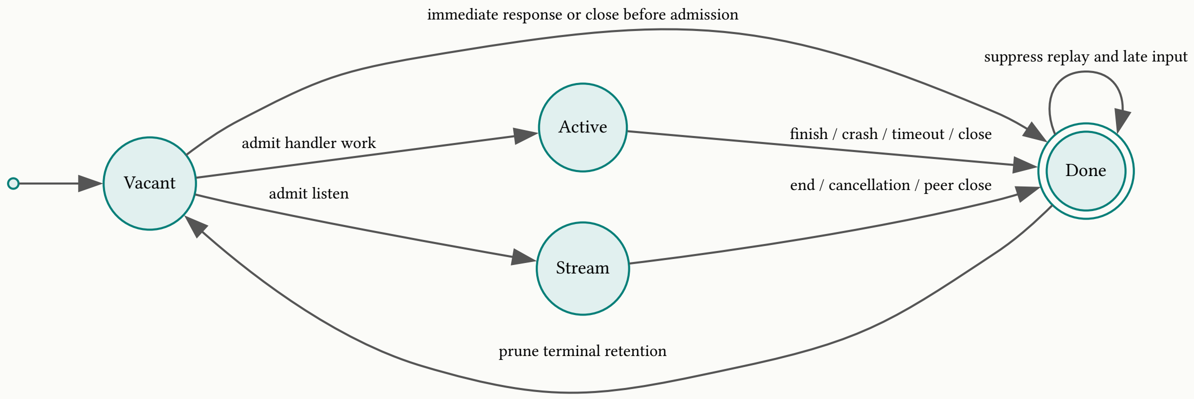
derived **Phase** · Exchange phase exchange-phase · exchange-life
Vacant means no retained record; Done is a retained terminal record.

derived **Routing** · RequestId × optional progress token × optional InvocationId
Active retains invocation and latest accepted progress; Stream retains a listen request.

RELATIONSHIPS

`1 : 0..1` One exchange owns at most one handler invocation or one subscription stream.

STATE MACHINE One response destination


immediate response or close before admission
suppress replay and late input
Active
admit handler work
finish / crash / timeout / close
Vacant
Done
admit listen
end / cancellation / peer close
Stream
prune terminal retention

Fresh identity is mandatory even after retention is pruned. A pruned id is not reusable authority.

Worker phase Pending Running Cancelling Exited

Invocation

entity · stateful · MCP · owned by runtime

One admitted native handler work item and its owned worker.

ATTRIBUTES

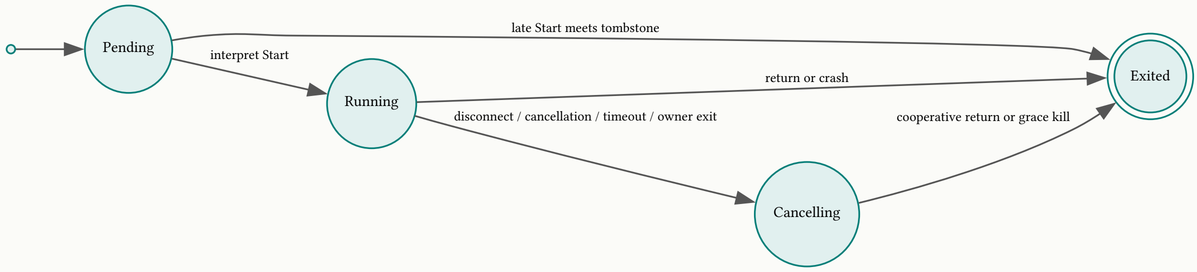
derived **Identity and work** · InvocationId × exchange × selected work × context
Retains method, name, metadata, arguments, input responses, and correlation without reusing the client’s id.

derived **Worker phase** · Worker phase worker-phase · worker-life
A terminal exchange may coexist with a Cancelling worker during its cleanup grace.

RELATIONSHIPS

`n : 1` Every invocation returns to exactly one runtime history and one retained exchange.

STATE MACHINE Handler execution and cleanup


Pending
late Start meets tombstone
interpret Start
Exited
return or crash
Running
disconnect / cancellation / timeout / owner exit
cooperative return or grace kill
Cancelling

Cancellation stops publication immediately; cleanup completion is a separate observation.

Server catalog

aggregate · stateful · MCP · owned by server description / live endpoint

The invariant owner of declaration and handler agreement.

ATTRIBUTES

authored **Services** · tools × resources × prompts × optional completion
Tools have unique names; listings retain registration order.

derived **Signing key and generation** · random key × monotonic registry generation
The immutable description retains the key across request runtimes; the endpoint owns generation changes.

RELATIONSHIPS

`1 : 0..n` Many request snapshots and streams refer to one current endpoint catalog.

Grant value-object immutable MCP owned by resource admission

Accepted endpoint authority over one resource and native principal.

ATTRIBUTES

derived **Principal and authority** · principal × ProtectedResource × List(Scope)
Actual accepted claims, including scopes beyond endpoint requirements, survive admission.

RELATIONSHIPS

`n : 1` An admitted HTTP request uses the endpoint’s configured Protection; arbitrary grant reuse is not a public protected-registry mechanism.

Products, sums and refinements	Representations	Preserved meaning
Request id	StringId / IntegerId	String 1 and integer 1 remain distinct; ids may repeat.
Reply / ToolResult	complete / input round; Succeeded / ToolFailed / InputRequired	Application refusal completes an exchange but does not prove no effect.
Structured discovery	Option(Value)	None is absence; Some(Null) is present null.
Content	text / image / audio / link / embedded resource	Bytes remain native; base64 appears only in the codec.
Application annotations	Option(Bool) hints and optional generic annotations	Omission differs from False; hints are untrusted policy evidence.
Failures	Reason × Correlation; Kind; Evidence	Opaque context does not change detailed reason matches.
Validated values	name 1–128 permitted ASCII; resource absolute HTTP(S) without fragment; nonempty scope	Admission validates the boundary; it does not normalize resource aliases or scope case.

- There is no persistent storage aggregate in Relay. Tombstones are node-local suppression records; receipts, durable idempotency, and retained business outcomes belong to applications. Commands and effects are detailed below; observation events report completed protocol transitions.

1.5 Protocol and schema boundaries

Revision codecs validate envelopes independently of application schema admission.

Modern wire codec

responsibility

Parse and classify revision-specific messages and encode exact results and errors.

interface

Internal JSON-RPC and v2026_07_28 admission/encoding functions; public protocol data in content, declarations, and tool input methods.

interactions

Receives bounded bytes; supplies admitted request metadata and checked Blueprint values to reducer dispatch.

invariants

Unambiguous protocol members, trustworthy ids, required metadata, matching routing headers, and distinct envelope families.

failure

Parse, invalid request, invalid parameters, unknown method, unsupported version, and routing disagreement remain distinct; no handler starts on refusal.

Contract	2026-07-28	Retained 2025-11-25 strategy
Readiness	Self-contained request metadata; discover optional for caller	initialize ⟶ AwaitingInitialized ⟶ initialized ⟶ Ready
HTTP	One POST; no session GET/DELETE or Mcp-Session-Id	POST JSON/SSE, GET notifications, DELETE termination, session registry
Headers	MCP-Protocol-Version, Mcp-Method, tool Mcp-Name agree with body	Negotiated revision and session-aware header validation
Result	Complete includes resultType, including isError results	Legacy result codec omits resultType
Cancellation	HTTP close; stdio notifications/cancelled	Revision-scoped restrictions including initialize cancellation
Logging and ping	Per-request logLevel metadata; no legacy setLevel or ping route	Legacy methods only where its schema requires them

- Modern `_meta` carries protocol version and configured client capabilities. Unsupported versions return the pinned −32022 outcome with supported revisions. Omitted tool arguments become an empty object; supplied non-objects fail before domain decoding.
- A notification has no ordinary JSON-RPC response. Accepted HTTP notifications receive 202 with an empty body; valid unsupported notifications and inbound responses must not become invocations. Parse errors omit an untrustworthy wire id while retaining the exchange destination.
- Modern cache fields are private with zero TTL in the implemented families. This does not establish cache revision or conditional ownership semantics. Required fields are validated against the frozen fixture rather than inherited from older SDK shapes.
- The **Protocol schema** in test fixtures is authority for wire shapes. A tool inputSchema/outputSchema is an application JSON Schema document represented as an exact Blueprint Value. Loading an **Input contract** admits Blueprint's finite profile; provider schema lowering belongs to LLM Wire adapters, outside Relay.
- Local input admission accepts Blueprint ObjectSchema and UnionSchema; OtherSchema requires an explicit object declaration; AnySchema fails closed. Output admission requires a schema document whose JSON representation is an object, not an object-shaped application output. This permits string and other structured output codecs.
- Remote declarations preserve schema documents and metadata before optional input_contract loading. A valid MCP declaration need not have a locally admitted Blueprint contract; unsupported features remain an explicit error rather than silently closing or weakening the schema. See **ADR-0001**, **ADR-0002**, and **ADR-0007**.

BEHAVIOR

Reject a mismatched modern request before dispatch

Given A POST claims a different version, method, or tool name in its routing headers and body.

When The server receives the MCP request.

Then The request receives the corresponding refusal without running its tool handler.

level: interface   area: Wire admission

BEHAVIOR

Retain complete tool failures as protocol results

Given A handler returns a safe application ToolError.

When The modern call finishes.

Then The client receives ToolFailed from an isError result with resultType complete.

Then Submission evidence is Completed; the application's effect outcome remains its own contract.

level: interface area: Result semantics

1.6 Definitions and service catalog

One retained contract drives declaration and dispatch while caller records remain native.

Tool construction and binding

responsibility

Admit a name and codec-derived declarations; bind native handlers before erasure.

interface

define / try_define; define_content / try_define_content; handle, handle_with_error_renderer, handle_call; independent metadata setters.

interactions

Blueprint owns schema/codecs; bound tools enter server.new or live register_tool; client.call reuses a Definition.

invariants

Input schema object root; output schema availability for structured tools; duplicate names rejected; no private generic errors disclosed by ordinary handle.

failure

Source-definition mistakes panic with the offending name; runtime admission returns DefineError; ordinary handler errors become a generic safe ToolError.

- Name admission permits ASCII letters, digits, period, underscore, hyphen and slash with length 1–128. Static server.new rejects duplicate names; runtime register_tool returns DuplicateTool. Unregistering an absent name is a no-op.
- with_input_schema publishes an explicit JSON object document in place of the codec-derived input schema. It does not replace the retained native codec, which still decodes every call. A non-object schema document is a source-definition panic; truthful agreement between an override and actual accepted inputs belongs to its author.
- handle accepts the caller's arbitrary error type and hides its details. The explicit renderer chooses safe content; handle_call receives Call(context) and returns complete, explicit content, metadata, or input-required requests. Context, progress, cancellation, client info, correlation and ids remain per invocation.
- A structured success encodes the native output and generates a text mirror unless explicit content replaces it. complete_with_meta replaces metadata on generated content blocks. Invalid output encoding and crashes become sanitized protocol failures; they never manufacture successful data.
- Annotations retain independent optional fields, including absent versus false. readOnlyHint and idempotentHint describe remote claims; applications choose whether to trust them when deciding uncertain-effect handling.

Service	Interface and interaction	Failure and invariant
Resources	static / template / try_template / template_with_matcher; list/read; native text/ blob contents	One simple variable per slash segment, including suffix/prefix; unsupported syntax requires explicit matcher; modern read failures return invalid params with the requested URI; private failure details remain hidden. See ADR-0009.
Prompts	prompt / prompt_call with named string arguments; list/get; role and content messages	Required arguments validated; private errors become invalid params; input rounds retain Call context.

Service	Interface and interaction	Failure and invariant
Completion	completion receives prompt/template reference, argument name/value and other arguments	Private error becomes internal error; output limited to 100 values with total and has_more.
Content	text/image/audio/resource link/embedded resource; annotations, icons, meta; text_of	Binary bytes preserved; text_of intentionally projects text only and joins text blocks.
Paging	Deterministic catalog order, signed family/offset cursor, client all-page traversal	Forged/cross-family cursors fail; authorized-view/revision binding remains a ruling.
Registry updates	server changes immutable description; endpoint/runtime register_tool changes live owner	Successful changes increment generation and notify active relevant streams; absent removals and failed additions do not invent changes.

BEHAVIOR

Modern resource failures retain the requested URI

Given A modern resource read names an unknown URI or its configured reader refuses the read.

When The server answers that read.

Then The caller receives invalid params with code −32602.

Then Error data.uri retains the requested URI.

Then Private reader failure details remain hidden.

level: interface area: Resource errors

BEHAVIOR

Re-evaluate visibility on named calls

Given A tool was listed for an earlier application context but is now hidden or not callable.

When The client names that tool in a new call.

Then The client receives the same inaccessible protocol outcome as an unknown name.

Then Neither argument decoding nor the handler runs.

level: interface area: Tool access

BEHAVIOR

Bind custom resource matching explicitly

Given A resource family uses syntax beyond one simple variable per segment.

When The application uses the built-in runtime template constructor.

Then It receives UnsupportedTemplateSyntax.

Then Using a caller matcher preserves the advertised template and the caller’s URI-recognition policy.

level: interface area: Service extension

- Current interfaces and rationale are recorded in `ADR-0002`. Access policy is per request; its evaluation is not atomic with an application’s later data changes or handler effects.

1.7 Reducer and runtime ownership

A pure state transition emits work; one actor decides when that work runs.

Reducer

responsibility

Maintain exchange, invocation, subscription, and catalog state for one authoritative history.

interface

init, step, perform; Received, Finished, Progressed, Crashed, TimedOut, ExchangeClosed, Notify, RegisterTool, UnregisterTool, EndStreams.

interactions

Produces Write, Start, Cancel, Close, and Admitted effects; perform runs outside step in the calling worker.

invariants

Every effect names an exchange or invocation; repeated admission and terminal input suppress output; progress requires the retained token and increasing nonnegative progress.

failure

Wire errors return immediate responses; unknown/late completions do nothing; copied state and invocations require an owner rather than granting exclusive execution.

Runtime actor

responsibility

Serialize reducer commands, account for live capacity, isolate handlers, and own cancellation cleanup.

interface

start / supervised, send_frame, exchange_closed, notify, register/unregister, close, stop; OutputWrite / OutputClose sink.

interactions

Unlinked monitored workers perform handlers; sink delivery is acknowledged; lifecycle telemetry reports transitions.

invariants

Duplicate exchange admission consumes no extra slot; bounded frames/depth/exchanges/tombstones; no late Start after cancellation tombstone; runtime remains alive until cancelled workers exit.

failure

FrameTooLarge, FrameTooDeep, TooManyLiveExchanges, RuntimeStopped are typed admission failures; worker crash/timeout is sanitized on its own exchange; sink error closes that exchange.

Runtime setting	Default	Boundary
Live exchanges	100	Admission refuses TooManyLiveExchanges before a new worker. Duplicate admission consumes no slot.
Frame / depth	1 MiB / 64	FrameTooLarge / FrameTooDeep before reducer admission.
Invocation / cancellation	30 s / 5 s grace	Timeout closes publication; grace bounds the runtime-owned handler's cleanup interval.
Invocation tombstones	60 s / 10,000	Late starts are suppressed within retained identity history; fresh ids remain mandatory after pruning.

Effect timing	Owner action	Observable guarantee
Received → admitted	Reduce bounded validated bytes with explicit context	Admitted observation follows protocol acceptance; refusal starts no worker.
Start	Retain worker ownership and start handler once	Invocation.started identifies execution; no exactly-once business-effect promise.
Progressed	Validate active token, monotonic progress; wait for sink acknowledgement	Backpressure reaches the producer; late or invalid reports are dropped.
Finished / crash / timeout	Produce one terminal output and close exchange	Private crash details stay off wire; later completions cannot answer again.

Effect timing	Owner action	Observable guarantee
Cancel	Signal handler selector; schedule grace kill	Wire output stops; handler may clean downstream work during grace.
Stop / owner exit	Cancel all active work and retain cleanup ownership	No unowned cancelled handler is declared cleaned before it exits or is killed.

- perform must run in the process addressed by signal_cancelled. Custom interpreters own scheduling, serialization, output acknowledgements, cancellation delivery and retention. A pure transition library cannot enforce these obligations for an arbitrary interpreter.
- Reducer closed exchange records are bounded at 10,000 and prune oldest records to three quarters when exceeded. Each runtime invocation **Tombstone** is retained for 60 seconds, at most 10,000 by default. Fresh node-unique identities prevent pruned records from aliasing new work.
- Every cancellation path signals first and preserves a default 5-second **Cancellation grace**, including timeout, close, stop, disconnect, stdio cancellation, and owner exit. The handler owns cancellation of effects it started elsewhere. Killing a worker is neither compensation nor proof a remote effect did not happen. See **ADR-0003**.

BEHAVIOR

A failed sink closes only its exchange

Given A runtime serves several exchanges and its output sink cannot write one exchange.

When That sink returns an error.

Then The affected exchange closes and its invocation receives cancellation.

Then Other exchanges retain their own destinations and lifetimes.

level: interface area: Runtime output

BEHAVIOR

A late completion cannot replace live work

Given An old exchange closed and a new exchange reused the same wire request id.

When The old invocation reports progress or completion.

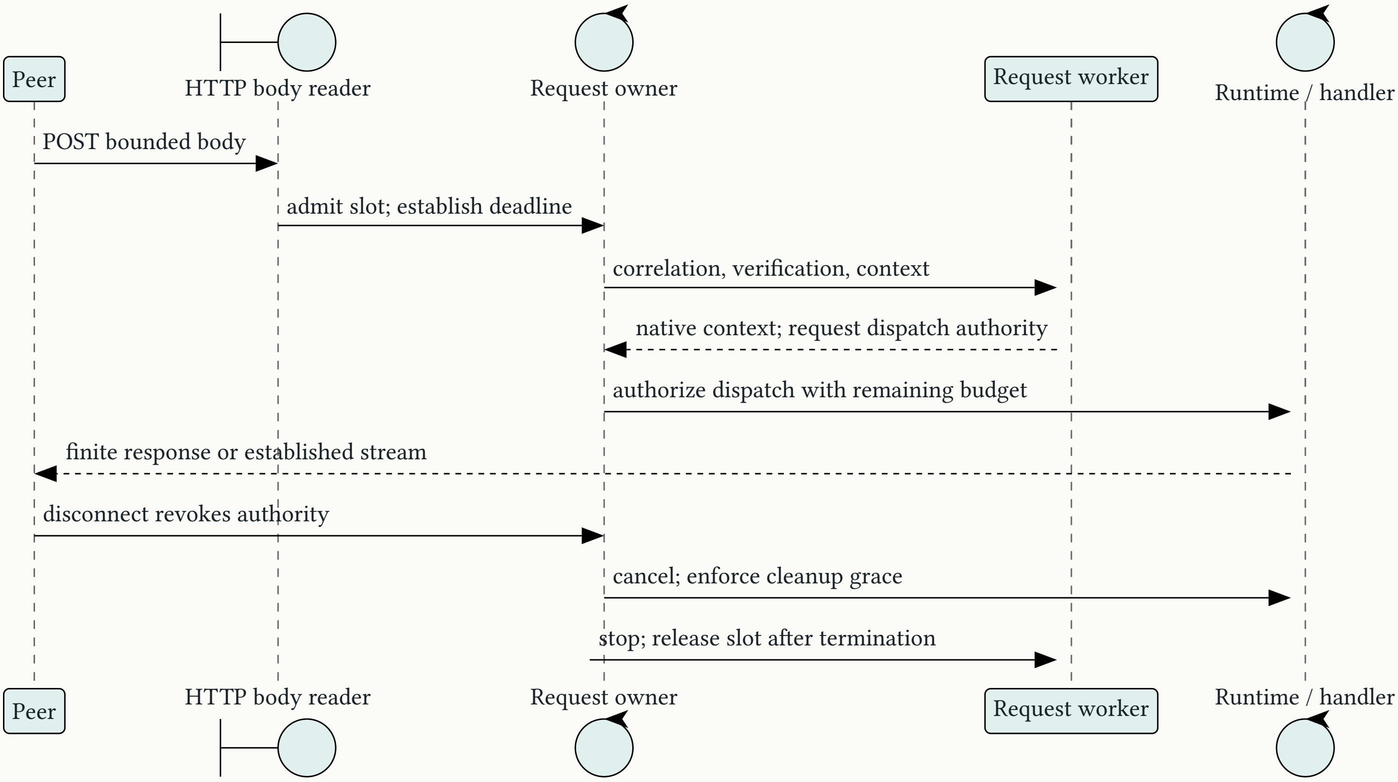
Then No output is written for the old invocation and the new exchange remains unaffected.

level: interface area: Runtime races

1.8 HTTP endpoint and limits

Relay owns the admitted request deadline and bounds cleanup separately.

SEQUENCE One deadline covers admission and execution


Peer
HTTP body reader
Request owner
Request worker
Runtime / handler
POST bounded body
admit slot; establish deadline
correlation, verification, context
native context; request dispatch authority
authorize dispatch with remaining budget
finite response or established stream
disconnect revokes authority
cancel; enforce cleanup grace
stop; release slot after termination

The request owner remains responsive while callbacks run. Mist body reads precede admission; established subscriptions retain their separate lifetime. Finite writes use the remaining deadline; owned writer termination precedes slot release. See `ADR-0010`.

Endpoint owns admitted work and cleanup

responsibility

Own current registry, admission slots, streams, optional listener and shutdown ordering.

interface

new / new_with_context / new_protected, setters, handler, handle, mist_handler, start, supervised, stop; live registration and notify.

interactions

Mist owns socket reads/writes; the endpoint leases a server snapshot and supervises each admitted request owner; the application router owns mounted endpoint lifetime.

invariants

Validate completed config before binding; loopback default; nonloopback unprotected bind requires explicit opt-in; header/media/Host/Origin validation and byte/depth limits precede protocol work.

failure

Typed startup settings/actor/listener failures; HTTP rejections with telemetry; capacity 503; body 413; buffered listen 406; pre-dispatch deadline expiry 504.

- Each **Request owner** remains responsive to deadline expiry, serving-process death, peer disconnect where observable, and endpoint shutdown while a managed worker runs application callbacks. The worker remains alive through the admitted exchange, including finite streaming output and any successfully established subscription.
- Slot admission establishes the **Admitted request deadline** before application correlation, context construction, or bearer verification. Admission consumes that budget; dispatch receives only the remaining duration. The owner checks deadline and cancellation authority before permitting dispatch and ignores late callback results. Actual handler entry rechecks authority after synchronous observations.

- Expiry before dispatch returns HTTP 504 while the connection remains usable. Disconnect suppresses the response. The runtime classifies invocation failures while the request owner accounts for their termination. Expiry after dispatch retains runtime cancellation and protocol outcome semantics; it does not prove application effects never ran.
- Expiry and disconnect begin a separate bounded cleanup interval using `Cancellation grace`. Normally completed buffered responses wait for owned work termination and acknowledged slot release within the original deadline. Retiring workers retain their request slots until owned request work terminates; capacity release never requires the callback to return voluntarily. Endpoint stop signals cancellation before listener shutdown and waits at most five seconds; it does not promise that every configured cleanup grace has elapsed before returning. Owner death also cancels owned request work; a scoped handler receives cancellation once even when its runtime is blocked in an observation.
- The deadline bounds finite response production and response waits. Mist registers its stream writer before the first pull; headers, events, keepalives and stream finalization check the current write budget. Writing the initial subscription acknowledgement successfully ends the finite request timer and uses the existing stream-capacity contract; the callback worker remains alive until that subscription closes. Mist's preceding body read remains outside this deadline.

Endpoint phase

Configured · Running · Stopping · Stopped · Failed

Endpoint lifetime

aggregate · stateful · MCP · owned by endpoint caller

The actor and optional linked listener that own admitted HTTP requests.

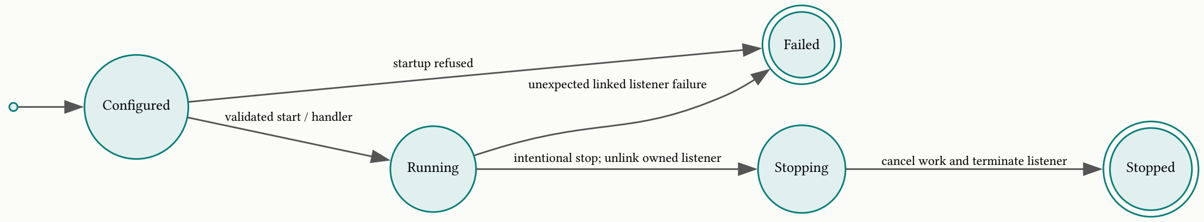
ATTRIBUTES

derived **Phase** · Endpoint phase endpoint-phase · endpoint-life
Running preserves unexpected listener failure propagation; intentional shutdown disconnects that link before termination.

RELATIONSHIPS

`1 : 0..1` A mounted endpoint has no Relay listener; a started endpoint owns one listener.

STATE MACHINE Listener ownership


Configured
startup refused
Failed
unexpected linked listener failure
validated start / handler
Running
intentional stop; unlink owned listener
Stopping
cancel work and terminate listener
Stopped

Unlinking a running listener or ignoring all linked exits would break ownership; `ADR-0006` *records the intentional-shutdown correction.*

Limit or policy	Default / refusal	Application point
Bind and protection	127.0.0.1, ephemeral; unprotected nonloopback start fails	Validated before listener startup; allow_unauthenticated is explicit caller policy.
Body and JSON depth	1 MiB ⟶ 413; depth 64 ⟶ 400	Mist bounded reader before slot; endpoint body check before context; depth before reducer parse. Mounted body allocation belongs to host.
Request slots and listen streams	1,024 and 64 ⟶ 503	Leased admitted requests/streams; does not count all incomplete network body readers.
Response budget	1 MiB response or cumulative stream event bytes	Broker enforces event output budget and acknowledged delivery; existing allocations are not erased by refusal.

Limit or policy	Default / refusal	Application point
Admitted request deadline	30 s	One absolute deadline starts at slot admission and covers callbacks, finite execution and response waits.
Cancellation and keepalive	5 s grace; 15 s SSE keepalive	Cleanup begins after cancellation; retiring work retains capacity until termination. Keepalive is transport liveness.
Host and Origin	Loopback names / bind host and configured origins	Case-insensitive header names; rejection precedes callback; missing Origin is permitted by current policy.

- handle consumes Request(BitArray) and returns buffered Response(BytesTree). It drops progress, cannot observe peer disconnect, and rejects subscriptions/listen with 406. mist_handler, start, and supervised retain streaming, keepalive and disconnect cancellation.
- Correlation, context, and verifier closures execute in the managed request worker, including buffered mounting. A callback-created Subject belongs to that worker and only its owner can receive from it; successful context construction does not terminate the worker. Long-lived clients, pools, and actors remain application-owned borrowed resources. Arbitrary detached or exit-trapping children and remote effects do not become Relay-owned resources.
- Modern MCP accepts POST only; protected-resource metadata has a separate GET path. No modern GET notification endpoint, DELETE session termination, or protocol session is added by this metadata route.
- Optional FFI helpers inspect Mist's Connection record to set a send timeout and observe buffered connection closure. They fall back on unfamiliar shapes; Mist upgrades require the real disconnect tests and record-contract check. See ADR-0003.

BEHAVIOR

Buffered mounting refuses listen streams

Given An application mounted the framework-neutral buffered handler.

When A client requests subscriptions/listen.

Then The response is 406; ordinary requests remain buffered and emit no progress.

level: interface area: HTTP mounting

BEHAVIOR

Blocked admission expires before dispatch

Given A configured callback remains blocked after its admitted request deadline.

When The deadline expires while the client remains connected.

Then The owner answers with a timeout refusal and revokes dispatch authority without waiting for the callback.

Then No callback result from that request may invoke a tool.

level: interface area: HTTP admission timing

BEHAVIOR

Admission consumes the execution budget

Given Admission has consumed part of an admitted request's original deadline budget.

When Its native context is accepted for dispatch.

Then Finite execution and response waits share only the remaining original budget.

level: interface area: HTTP admission timing

BEHAVIOR

Delayed policy cannot start expired work

Given Context construction succeeds before the deadline but tool admission policy remains blocked.

When The policy returns after expiry or cancellation.

Then The invocation gate rejects the expired request before starting its handler.

Then Queued admission results cannot restore dispatch authority.

level: interface area: HTTP admission timing

BEHAVIOR

Disconnect closes blocked admission authority

Given A streaming-capable client connection closes while its callback is blocked.

When The request owner observes that closure.

Then It suppresses output, revokes dispatch authority, and begins bounded cleanup.

Then Late admission success cannot start a tool invocation.

level: interface area: HTTP cancellation

BEHAVIOR

Retiring work remains capacity bounded

Given All request slots are held by active or retiring owned work.

When Another request attempts admission.

Then The server answers with an overload refusal until a slot's owned work terminates.

Then Cleanup expiry forces termination without waiting for the callback to return.

level: interface area: HTTP cleanup

BEHAVIOR

Endpoint shutdown owns admitted callback work

Given A server has a request blocked in context creation or verification.

When The server stops or the serving owner dies.

Then The request loses dispatch authority and its owned work terminates within cancellation grace.

Then Borrowed application resources retain their application-owned lifetime.

level: interface area: HTTP cleanup

BEHAVIOR

Callback failure isolates the mounted caller

Given An application invokes the buffered mount from a live caller process.

When The managed callback crashes.

Then The request answers with a server-error refusal and its owned work is cleaned up.

Then The caller remains alive and the server can admit subsequent requests.

level: interface area: HTTP callback ownership

BEHAVIOR

Callback resources retain the request worker

Given A callback creates a Subject in its managed request worker.

When Context creation succeeds and finite execution begins.

Then The Subject’s owner remains alive through the admitted exchange.

Then The Subject still permits receiving only in its owning process.

level: interface   area: HTTP callback ownership

1.9 Subscriptions and live registry changes

Acknowledgement follows successful owner registration, while churn history stays unretained.

- Each **Subscription** has an independent owner and cancellation destination. A listing **Cursor** selects a position, while endpoint generation orders live catalog reconciliation. Neither is business identity.

Endpoint stream registration

responsibility

Coordinate current catalog, subscription capability/filter and publication of the initial stream frames.

interface

subscriptions.listen through reducer, endpoint stream attach/reconcile, notify, register/unregister and end_streams.

interactions

The lease carries server snapshot and generation; the endpoint serializes activation and reconciles changed snapshots into request runtimes before acknowledgement release.

invariants

Same-name replacement updates handler and metadata; no lifetime change journal; no acknowledged ghost stream; unsupported notification kinds are omitted.

failure

Owner death, timeout, failed reconciliation or writer failure tears down runtime/broker/stream and queues unregister if admission may have happened.

Subscription phase Pending Active Closed

Subscription

entity   stateful   MCP   owned by endpoint/runtime and client stream owner

An acknowledged notification filter tied to one listen exchange.

ATTRIBUTES

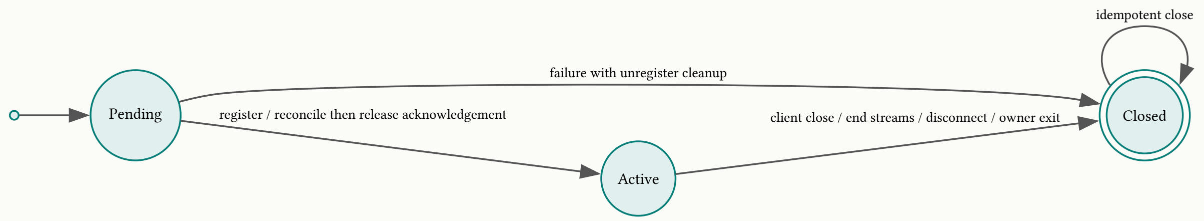
derived **Phase** · Subscription phase subscription-phase · subscription-life
Pending bytes are held until owner registration succeeds.

authored **Filter** · resource-updated URI / tools/resource/prompt list-change kinds
The acknowledgement retains only offered kinds, including tools-list change for an empty dynamic registry.

RELATIONSHIPS

`n : 1` Many subscriptions share one stdio child or current HTTP endpoint catalog.

STATE MACHINE No acknowledgement before activation


idempotent close
failure with unregister cleanup
Pending
register / reconcile then release acknowledgement
client close / end streams / disconnect / owner exit
Closed
Active

- Every accepted catalog mutation increments generation. A changed snapshot reconciles the complete current catalog and produces a tools-list-change notification before activation acknowledgement; reconciliation cost depends on snapshot/current size rather than prior churn.
- Stdio subscription ids correlate events within the child owner. FIFO queues preserve notifications for other subscriptions across a timed-out read; closing one subscription sends its cancellation, marks it closed and discards only its events. The shared child remains usable.
- Pending stdio frames and notifications have a 256-frame bound; overflow closes the owned child before evicting its handle. This is a retained-data count bound, alongside frame byte bounds, rather than a universal allocation or stream-delivery guarantee. See ADR-0006.

BEHAVIOR

Same-name replacement becomes current before acknowledgement

Given A pending stream’s catalog snapshot contains a tool removed and re-registered under the same name.

When The server activates that stream.

Then The stream observes tools-list change and subsequent requests use replacement metadata and handler.

level: interface area: Registry races

BEHAVIOR

Timeout preserves another stdio subscription's events

Given Events for subscription B arrive while a read waits on subscription A.

When A’s read times out.

Then B’s events remain available in arrival order and the child remains owned.

level: interface area: Subscription reads

1.10 Stdio transport

A trusted parent owns the child boundary and one actor serializes pipe traffic.

Server stdio adapter

responsibility

Frame bounded newline JSON-RPC and serialize stdout without nonprotocol noise.

interface

serve blocks main; serve_with config sets chunk size and runtime limits.

interactions

Binary stdin reader, incremental framer, serialized writer and runtime; EOF and asynchronous writer error reach the serving owner.

invariants

4 KiB reads by default; LF or CRLF framing; UTF-8 may split between byte chunks; stdout exclusively protocol; oversized tail discarded until newline.

failure

Invalid settings, ReadFailed, StdoutBroken, StartupFailed and InvalidTrailingData are typed; broken stdout terminates even with idle stdin; shutdown retains cleanup ownership.

Client child owner

responsibility

Launch executable directly, own its port, and serialize requests and subscription reads.

interface

client.stdio config / connect; internal request, subscribe, next and cancellation protocol.

interactions

One actor retains child, framer, FIFO frames/notifications and closed subscription ids; outer client polls cancellation and budgets.

invariants

No shell fallback; command closure avoids accidental inspection; stderr isolated; reply ids correlated; unrelated responses fail; pending and buffered limits enforced.

failure

Child exit, malformed frame, timeout, cancellation, client close, oversize and overload become typed client reasons/evidence; terminal errors close port before clearing state.

- Stdio provides no bearer authorization. Its trust boundary is the parent process and launched executable; metadata or a caller-supplied principal does not supply transport authentication.
- Cancelling one active stdio call sends notifications/cancelled. Closing the client closes its owned child and in-flight waits; repeated or concurrent close observes the owner via a monitor rather than treating dead-owner acknowledgement as success work.
- EOF in the middle of a nonterminated frame returns trailing-data failure. Idle broken output and child/owner death require asynchronous lifecycle tests; line parsing alone cannot prove cleanup.
- Legacy retained stdio shutdown and reconnect behaviors require their own strategy and tests. Existing bounded frames and child ownership do not establish legacy session conformance. See **ADR-0003** and **ADR-0007**.

BEHAVIOR

Stdio overflow closes its child before eviction

Given A child sends more retained frames than the client permits.

When The owner detects overflow.

Then The child is closed and subsequent calls fail as a terminal connection.

level: interface area: Child lifetime

1.11 Client calls and submission evidence

Client views share a connection while deadlines, correlation and cancellation remain explicit.

Client configuration and connection

responsibility

Validate target/settings, own resources it creates, and keep caller-owned HTTP resources distinct.

interface

http(url), stdio(executable,args), in_process; pure setters; connect; close; with_http_client injection.

interactions

HTTP Gun provides transport, TLS, pool and cancellation; stdio owns child; in-process owns runtime; immutable views share the peer.

invariants

Validation precedes connection; plaintext headers off loopback require explicit opt-in; absolute HTTP(S) URL excludes userinfo/query/fragment and invalid escapes; no automatic retry.

failure

InvalidConfig, ConnectFailed and Refused occur before request delivery; setup errors have a fresh correlation no server saw.

Calls, lists and continuations

responsibility

Encode arguments, correlate responses, preserve typed protocol outcomes, bound traversal and retain originating continuation.

interface

call, call_discovered, discover, list services, read_resource, get_prompt, complete, listen/next/close_subscription, resume; list_raw / call_raw; per-client deadline/cancellation/correlation/key views.

interactions

Definition output codec decodes native values; discovered calls preserve optional exact structured Value and media; request metadata advertises only configured input methods.

invariants

One response id must match request; response/version/size validation; each continuation retains original client/name/arguments/state; supplied responses cover every request key once with object values.

failure

Opaque Error retains Reason and Correlation; stable Kind/Evidence classify without diagnostic parsing; malformed or oversized response does not establish application outcome.

HTTP client request ownership

responsibility

Bound dynamic header acquisition and transport under one request budget while preserving caller-owned HTTP bodies.

interface

with_headers retains its callback signature; request timeout and per-call deadline/cancellation apply before dispatch.

interactions

A request-owned worker computes headers. A responsive owner watches caller death, owned-peer closure and cancellation. The calling process opens and reads HTTP Gun bodies.

invariants

Each wire request starts one relative budget before header acquisition. A supplied absolute deadline can shorten the budget. Header work ends before dispatch; timeout or cancellation prevents late dispatch. HTTP Gun submission evidence remains authoritative after dispatch.

failure

Header callback failure becomes InvalidConfig(Headers) without exception details. Pre-dispatch timeout/cancellation carries NotSent. Owned work terminates on result, cancellation or caller death.

- A Relay-owned HTTP peer retains a closure signal until explicit close or connector exit. Request owners monitor that signal. Injected HTTP clients have no Relay-owned closure signal.
- **Header acquisition** runs header callbacks in a temporary request-owned process. They return header data, not process-affine resources. Application resources captured by the callback remain borrowed; arbitrary detached children and remote effects are not adopted. See **ADR-0011**.
- Subscription opening uses the same budget for headers, HTTP opening and the acknowledgement. Successful acknowledgement removes the relative opening limit. The explicit absolute deadline and cancellation continue to bound the established subscription. Closing or failing the stream releases its request owner.
- An injected HTTP Gun client's existing absolute deadline remains intact unless the caller explicitly installs a Relay deadline view. A relative Relay timeout never replaces that HTTP deadline. Per-page request budgets and separate continuation requests retain their existing boundaries.

BEHAVIOR

Blocked client headers cannot outlive their request

Given A dynamic header callback has entered and has not returned.

When The request deadline expires or its cancellation is observed.

Then The caller receives a typed NotSent failure, the callback terminates, and its eventual release cannot dispatch the request.

level: interface area: HTTP client lifetime

BEHAVIOR

Subscription acknowledgement ends only the opening budget

Given The client opens a subscription within its request budget.

When The subscription acknowledgement succeeds.

Then Notifications remain readable beyond the relative opening budget while explicit deadlines, cancellation and caller ownership remain effective.

level: interface area: HTTP client lifetime

Client setting	Default	Boundary
Connect / request timeout	10 s / 30 s	Connection startup and individual request; absolute deadline may shorten request budget.
Response bytes	1 MiB	ResponseTooLarge retains delivery evidence rather than manufacturing a tool outcome.
All-page listings	256 pages / 10,000 items	ListingLimitExceeded; a repeated cursor is MalformedResponse.
Stdio pending calls	64	TooManyPendingCalls before enqueueing additional work.
Advertised input methods	None	Explicit configuration opts into Elicitation, Sampling or Roots.

Control / result	Contract	Decision left to caller
Timeout and deadline	30 s per request; absolute deadline view shares a total budget and may shorten it	Multi-page traversal otherwise resets request timeout per page; choose total budget explicitly.
Cancellation	HTTP closes that request connection; stdio sends notification; in-process signals handler	Handler cleanup and external effect reconciliation remain separate.
Client close	Owned peer cancels in-flight calls; injected HTTP Gun remains running and calls may finish	Use per-call cancellation to stop work on the shared HTTP client.
ToolResult	Succeeded(output,content), ToolFailed(content,structured), InputRequired(requests,state,continuation)	Continue interaction or choose lossy output projection.
NotSent	Configuration, pre-send refusal, invalid input, overload; no request sent	Retry only within caller policy and remaining budget.
MaybeSent	Lost/oversized/malformed response or in-flight timeout/cancellation	Mutating work may have happened; use principal+key receipts or reconciliation.
Completed	HTTP status / RPC answer / tool result / input request observed	Completion is exchange evidence; tool failure can follow effects.

Control / result	Contract	Decision left to caller
Output projection	require retains transport error and classifies refusal/needs-input; require_discovered uses actual presence	InputRequired projection discards the continuation deliberately; text projection discards nontext media.

Detailed Reason	Kind	Evidence / retained payload
InvalidConfig	Configuration	NotSent; invalid ConfigField.
ConnectFailed / Refused	Unreachable	NotSent; failed establishment / HTTP destination policy.
TimedOut / Cancelled / ConnectionClosed	Timeout / Cancellation / Unreachable	Explicit Evidence argument records whether submission may have occurred.
TooManyPendingCalls	Overloaded	NotSent; configured pending-call limit.
ResponseTooLarge	TooLarge	MaybeSent; response byte limit.
HttpStatus	Rejected	Completed; status and optional WWW-Authenticate challenge.
RpcError	Protocol	Completed; code, message and optional exact data.
MalformedResponse	Protocol	MaybeSent; diagnostic detail for logs.
UnsupportedVersion / UnsupportedInputRequest	Protocol	Completed; supported revisions / unsupported requested method.
InvalidArguments / InvalidInputResponses	InvalidInput	NotSent; arguments diagnostic / exact one-object-per-request-key failure.
ListingLimitExceeded	TooLarge	Completed; page or item limit encountered after a listing response.

- list_raw preserves exact item values while retaining the same page/item bounds and repeated-cursor protection. call_raw returns the exact result value through the same request validation, evidence and controls; tools/call, prompts/get and resources/read require a routing name, while other methods reject one. Raw access does not bypass revision or transport admission.
- Error Kind remains closed: Configuration, Unreachable, Timeout, Cancellation, Overloaded, Rejected, Protocol, InvalidInput, TooLarge. Detailed Reason includes status and WWW-Authenticate, RPC code/message/data, version, input and limit detail, and delivery evidence. Log describe_error with error_correlation; never branch on prose.
- is_retryable is an advisory classification, not a retry loop or effect proof. NotSent unreachable/timeout and overload can be retried; MaybeSent unreachable/timeout requires application-approved idempotence; 429/503 retain explicit HTTP status. Read-only hints from untrusted peers do not supply that approval.
- Input methods Elicitation, Sampling and Roots are configured explicitly; no responders are advertised by default. Handler requests are filtered to client capability; unsupported incoming methods fail instead of automatically executing local capabilities. A **Continuation** retains its origin rather than retargeting subsequent answers.
- Connection copies share lifetime. Views attach an absolute deadline, cancellation token, correlation or idempotency key without creating resources. Borrowed HTTP Gun correlation can supply the base view; invalid wire correlation stays local.
- Absent discovered structuredContent is None, including continuation rounds; explicit null is Some(Null). output.require_discovered projects text only on actual absence and does not inspect outputSchema to invent presence. See **ADR-0005**.

BEHAVIOR

Invalid idempotency key fails before sending

Given A client view has an empty, oversized or non-visible-ASCII idempotency key.

When It attempts a request.

Then The result is InvalidArguments with NotSent evidence and the call’s correlation.

level: interface area: Client controls

BEHAVIOR

Preserve present structured null

Given A discovered call completes with structuredContent explicitly null and nonempty text.

When The caller requires the discovered output.

Then The answer remains Null rather than the content’s text.

level: interface area: Discovered output

1.12 Configuration surface

Every transport read, wait and retained queue has an explicit bound with a named setter.

- Callback execution and host body allocation have separate owners, described in the HTTP and client sections. A mounted host allocates its own request body before Relay sees it.

Setting	Default	Change with
HTTP bind	`127.0.0.1`, ephemeral port	`http.with_bind`
HTTP bind off loopback without protection	`start` fails	`http.new_protected` or `http.allow_unauthenticated`
Host and Origin allow-lists	Loopback names and the bind host	`http.with_allowed_hosts`, `http.with_allowed_origins`
HTTP request body	1 MiB, then 413	`http.with_max_body_bytes`
HTTP response, or one stream’s events	1 MiB	`http.with_max_response_bytes`
HTTP admitted-request deadline	30 s from slot admission	`http.with_request_timeout`
SSE keepalive	15 s	`http.with_sse_keepalive`
Concurrent HTTP requests	1,024, then 503	`http.with_max_concurrent_requests`
Concurrent `subscriptions/listen` streams	64, then 503	`http.with_max_listen_streams`
JSON nesting depth	64, then 400	`http.with_max_json_depth`, `runtime.with_max_json_depth`
Runtime live exchanges, frame size	100; 1 MiB	`runtime.with_max_live_exchanges`, `runtime.with_max_frame_bytes`
Runtime handler timeout	30 s	`runtime.with_invocation_timeout`
Cancelled handler grace before kill	5 s	`runtime.with_cancellation_grace`
Tombstones	60 s, at most 10,000	`runtime.with_tombstone_retention`, `runtime.with_max_tombstones`
Closed exchange records per connection	About 10,000	Fixed
Stdio read chunk	4 KiB	`stdio.with_chunk_size`
Client request timeout	30 s	`client.with_timeout`; per call `client.with_deadline`
Client connect	10 s	`client.with_connect_timeout`
Client response size	1 MiB	`client.with_max_response_bytes`
Client listings	256 pages, 10,000 items	`client.with_listing_limits`
Client stdio pending calls	64, then `TooManyPendingCalls`	`client.with_max_pending_calls`
Client input methods advertised	None	`client.with_input_methods`
Client headers over plain HTTP off loopback	Refused	`client.allow_plaintext_headers`
Client retries	None; decide with `client.evidence`	Not configurable
Request correlation	The client’s, else a fresh one	`http.with_correlation`, `client.with_correlation`
Request idempotency key	None	`client.with_idempotency_key`

Setting	Default	Change with
Completion values per response	100	`completion.Values(total:, has_more:)`

1.13 Bearer admission and tool policy

A trusted verifier supplies claims; Relay admits resource authority and applies current request policy.

- The configured verifier's **Attestation** is admitted against **Protection** to produce a **Grant**. The caller translates that grant into **Application context** it does not replace endpoint admission with a web-login identity.

Resource-server authorization

responsibility

Parse bearer headers, admit exact resource/scopes, render challenges and metadata.

interface

BearerToken/ProtectedResource/Scope constructors; verifier(name, fn(token,correlation)); attestation; admit; Grant readers; Protection; challenge and resource_metadata.

interactions

new_protected calls the selected verifier and builds native context from Grant; Hellhound may supply an independently configured verifier recipe but Relay does not import it.

invariants

Singleton exact resource audience; all endpoint scopes present; retain actual scopes; correlation never affects admission; token not retained as Grant field.

failure

MissingToken, VerificationFailed, ResourceNotGranted and MissingEndpointScope produce distinct internal decisions; public HTTP status/challenge is sanitized.

Refusal	Response	Caller action
Missing token	401 Bearer with resource_metadata; no error code	Discover protected-resource metadata or supply credentials.
Rejected/unmapped/wrong-resource token	401 invalid_token	Refresh or reauthorize under resource/issuer policy; no infinite 401 loop.
Missing required scope	403 insufficient_scope and required scopes	Explicit step-up policy.
Verifier unavailable	503 without authentication challenge	Verifier availability is not invalid identity.
Hidden/denied/unknown tool	One inaccessible protocol outcome	No tool-existence disclosure; handler and decoder untouched.

- The **Verifier** owns signature, issuer, expiry, status, algorithm, key rotation, introspection and sender-constraint checks. Its Attestation constructor can assert arbitrary claims; the application selects and trusts that implementation. Empty audience/scope strings are filtered before exact admission; duplicate/additional audiences still fail the singleton profile.
- Protection stores an absolute HTTP(S) resource URI without fragment, nonempty exact scopes and declared authorization servers. URI and scope equality are exact configured strings; no path, case, alias or audience-set normalization grants authority.
- BearerToken captures raw value inside a closure to reduce accidental inspect disclosure; token_value intentionally reveals it to the verifier. A caller-owned principal can still retain secrets, so token redaction remains an adapter and observation obligation.
- new_protected runs admission on each HTTP request before building context. Grants are translated into native context and server.with_tool_access filters listings and rechecks visible+callable before argument decode. Per-input permissions remain in typed handlers.
- The metadata GET at `/.well-known/oauth-protected-resource` plus resource path reports resource, authorization servers, scopes and header bearer method. Challenge header escaping and status behavior belong to Relay; token validity remains outside it.

- A Hellhound web identity cannot substitute for a Relay grant. Relating a login identity to bearer principal, tenant and business access is explicit application policy. Multiple trust domains sharing resource and principal type need explicit verifier-origin reuse rules. See **ADR-0004**.

BEHAVIOR

Additional audience prevents admission

Given A trusted verifier returns a valid token attestation with the configured audience and one additional audience.

When Relay admits it under the singleton resource profile.

Then It returns ResourceNotGranted and no Grant or tool work is admitted.

level: interface area: Resource grants

BEHAVIOR

Unavailable verification is not invalid authentication

Given The configured verifier cannot obtain enough evidence to decide token validity.

When It returns VerifierUnavailable.

Then The protected server answers 503 without WWW-Authenticate and emits the unavailable decision.

level: interface area: Authorization failure

1.14 Composition ports and observations

Applications translate protocol evidence and native execution outcomes at small explicit boundaries.

- The application maps **Submission evidence** into its effect model. **Correlation** joins observations, while **Idempotency key** acquires business meaning only under authenticated principal and application policy.

Typed tool output port

contract

Expose answer, failure classification, evidence and reply metadata without importing an agent runtime.

responsibility

Return native output or structured projection failures.

interface

relay/client/output require, require_discovered, error_kind, evidence, describe_error, meta; tool.complete_with_meta; content.text_of.

interactions

An application translates its durable-run or agent vocabulary; full ToolResult remains available for media and interaction.

invariants

CallFailed retains original Error; ToolFailed and InputRequired retain Completed evidence; metadata lookup imposes no Fabric key.

failure

Projection loss is explicit; uncertain transport does not become application refusal.

- Fabric invoke owns principal+key run identity, matching-input reopen, wait, cancellation and durable outcome. A Relay recipe copies request correlation/key, builds invoke request from native principal/input, sets its wait shorter than Relay's invocation budget, and returns native result plus application run metadata.
- For an unkeyed request the recipe forwards tool.cancelled to the owned invoke wait; keyed work may retain its durable run under application policy. Relay's JSON-RPC id and InvocationId never identify that durable run.
- Discovery has separate listing and schema/name admission boundaries. The application decides peer trust before treating readOnlyHint as evidence that MaybeSent work is safe to repeat. tool_hub trusts the inventory endpoint it owns; this is not a general trust policy for arbitrary remote peers.
- Secure MCP uses a Hellhound resource-verifier recipe and native principal context. Hellhound validates token cryptography/introspection with its own limits; Relay maps resource/scope admission and challenges. Relay's request deadline bounds verifier and context callbacks independently of Hellhound's own limits.

Correlation and telemetry

responsibility

Carry request observation identity and emit typed protocol facts after transitions.

interface

client.with_correlation, http.with_correlation, tool.correlation, telemetry event descriptors; Sinal observers.

interactions

HTTP x-correlation-id and `_meta` io.github.gleam-dream/correlation join client, endpoint, verifier, handler and downstream events.

invariants

Every request gets a Correlation; precedence is transport override, valid carried value, then fresh; accepted wire value is 1–128 visible ASCII; correlation never authorizes.

failure

Invalid carried correlation is ignored rather than refusing the call; before-call errors carry a fresh local-only correlation; rejected frame has no admitted request correlation.

Event	Transition observed
frame.rejected / http.rejected	Boundary refusal before admitted work
request.admitted	Accepted protocol request
invocation.started / completed / cancelled / crashed	Worker execution, result, cancellation intent or crash/ timeout; duration on completion
exchange.closed	Terminal destination with its request correlation
authorization.decided	Configured verifier/resource admission decision
client.call	Call outcome and elapsed duration with outgoing correlation

- A handler reads the JSON-RPC id with `tool.request_id`, which returns the id as the client sent it, may repeat across clients and is fresh on a retry. `tool.invocation_id` is Relay's own id, new for every request, and joins the call's telemetry. Neither identifies a durable operation.
- Observation callbacks do not choose protocol results. Request ids, correlation and idempotency keys are untrusted independent values; the authenticated principal controls business authority. The optional idempotency metadata key is io.github.gleam-dream/idempotency-key, accepted only for 1–128 visible ASCII. See ADR-0005.

1.15 Validation and extension ownership

Structural validity, runtime fidelity and complete conformance require different evidence.

Public testing and evidence tools

responsibility

Provide deterministic local serving/calling and validate compiler, schema, runtime and external contracts independently.

interface

testing.connect/error/request/body_text/verifier/unavailable_verifier; test suites and peer fixtures; schema/negative/checksum scripts; pinned conformance runner.

interactions

In-process uses the same wire and runtime; fixture peers exercise hostile declarations/continuations/subscriptions; conformance launches a local endpoint; external recipes compile in Fabric consumers and application checkouts.

invariants

Schema checksum and sibling revisions recorded; tests use actual encoder bytes; unsupported features and suite omissions remain explicit; no selected scenario or expected-failure file claims full conformance.

failure

Wrong native handler codec fails compilation; schema mutations fail validation; runtime races need process/socket assertions; client and auth suites require independent invocation.

Evidence layer	What it proves	What it cannot prove
Compiler / negative fixture	Native codec/handler agreement and public construction boundaries	Runtime lifecycle or token authenticity
Frozen JSON schema corpus	Actual wire output shape and required-field mutations	Behavior, framing or complete protocol conformance
Pure/property/race histories	One authoritative transition history, ids, terminal suppression	Exclusive interpreter ownership or stopped external effects
Runtime / real HTTP / child tests	Measured process cleanup, acknowledged backpressure, disconnect and FIFO behavior	All OS/library versions or untested admission/allocation bounds
Pinned server conformance	Emitted assertions for frozen 2026-07-28 server scenarios	Client/auth/legacy support or zero-check scenario coverage
Compiled usage examples	Each documented recipe builds and runs against the public API from a separate package, with caller-native types and failure handling	General third-party schema/peer trust or production provider acceptance

- CI invokes Nix/treefmt and the pinned design-layer checker directly. The local scripts/test.sh compiles Gleam and authored Erlang with warnings as errors and runs application tests, compiler fixtures, schema comparisons and the server suite. Local source-policy checkers, profile registries and provenance receipts remain removed. See ADR-0015.
- Generated dependencies, TLS fixtures, the pnpm lockfile and deliberate negative compiler fixtures are excluded from authored formatting. No benchmark harness or latency claim exists.
- The server check runs the pinned conformance package against all server scenarios of spec 2026-07-28 and retains structured results and the local server log on success or failure. The upstream CLI reports failed assertions with a nonzero exit status. Scenario inventory and warning policy are not separately enforced; emitted assertions establish only their tested scope.
- Each **Sibling dependency** of Relay (Sinal, HTTP Gun and JSON Blueprint) uses a Hex version requirement. Each manifest records the resolved versions and checksums; dependency updates are explicit. CI fetches the dependencies like any other and persists no credentials; fork pull requests run the same jobs. A change that spans Relay and a sibling is verified by a whole-graph local-source capture rather than a path override of Relay alone. See ADR-0016.
- The MCP schema is fetched lazily by the comparison through Nix at its pinned commit and verified against its recorded checksum. Nix caches it in the store; entering the development shell, formatting and ordinary compilation do not fetch it, and no copy is committed. scripts/conformance retains package and lockfile; acceptance uses the pinned 0.2.0-alpha.10 requirement anchor. The reference pins, comparison classification and limits are kept in the oracle document. See ADR-0007.
- Usage is kept as compiled, runnable recipes in the usage examples package, one module per recipe. Shared pinned Snips synchronizes root README excerpts from named source regions; no local custom README checker is retained. The application test script compiles and tests the package, so an API change that breaks a recipe fails validation. No standalone usage guide exists. See ADR-0014.
- Pinned **Conformance evidence** covers assertions emitted by the external suite; failed assertions fail the integration test. The unpublished fixture validates its requested elicitation response, mutates the live tool catalog and publishes prompt notifications through the endpoint owner. See ADR-0009.
- Keep split UTF-8/large frames, broken stdout with idle stdin, owner/child death, equal wire ids, duplicate admission, cancellation races, tombstones, held-writer mailbox bounds, burst disconnect, registry-generation barriers, FIFO subscription timeout and overflow fixtures. Source assertion names supply reproduction points, not a test-run diary.
- Revision codecs, custom template matcher, generic verifier and reducer interpreter are extension contracts with different ownership. A new transport must prove framing, limits, disconnect/cancellation and cleanup from an external consumer; a buffering interface cannot fabricate socket ownership.
- A new protocol revision carries its schema, requirements, supported-method set and independent lifecycle strategy. Supported-version removal requires a major version; deprecation lasts at least a minor release. Older than 2025-03-26 is excluded.
- Release verification retains Erlang/OTP target matrices, pinned TypeScript/Python peers in both directions, Inspector smoke, fuzz/soak/cancellation storms, bounded process/mailbox evidence and continuous latest-harness/schema drift. No completed hardening run is inferred from the standing design.

1.16 Retained protocol strategies

Full scope includes separately modeled compatibility and authorization flows.

- Each **Protocol strategy** shares native definitions and ports with the others and keeps its own codec and lifecycle. The modern per-request strategy is implemented; the retained strategies below are designed scope.

Legacy compatibility strategy

responsibility

Admit a session through initialize, negotiate revision/capabilities and gate requests on initialized readiness.

interface

Intended revision-bound client/server codecs and session owner; no legacy public runtime is implemented.

interactions

Share native definitions, results and application ports; keep session registry, idle/max-session limits, transport resumption and revision notifications separate from modern POST-only execution.

invariants

Reject repeated initialize, stale/unknown session, early normal calls and post-initialization version disagreement; resultType remains absent.

failure

Revision-specific protocol failures, cleanup, unsupported capability and session termination outcomes; initialize cancellation rules must match the frozen revision.

Legacy session phase

AwaitingInitialize AwaitingInitialized Ready Closed

Legacy session

aggregate **stateful** **MCP** **owned by legacy session runtime**

Retained design for negotiated 2025-11-25 lifecycle, absent from modern state.

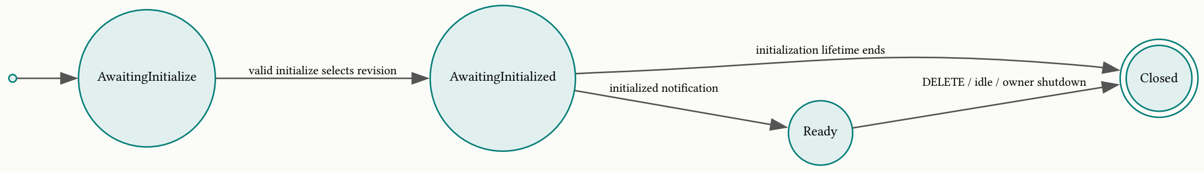
ATTRIBUTES

derived **Phase** · Legacy session phase legacy-phase · legacy-life
Only valid initialize establishes negotiated version, capabilities and session identity.

RELATIONSHIPS

`1 : 0..n` One ready legacy session may own requests and its notification/resumption stream.

STATE MACHINE Retained legacy handshake


AwaitingInitialize
valid initialize selects revision
AwaitingInitialized
initialization lifetime ends
initialized notification
DELETE / idle / owner shutdown
Ready
Closed

No transition from modern discovery creates this session. Early or repeated handshake messages reject without changing the phase.

MCP client authorization strategy

responsibility

Own MCP-specific resource discovery, issuer-bound credentials, scope step-up and bounded reauthorization.

interface

Intended static bearer, authorization-code PKCE, client credentials and private-key-JWT modes; preregistration, permitted dynamic registration and client metadata documents; pluggable memory/file token custody.

interactions

Challenge ⟶ RFC 9728 resource metadata ⟶ RFC 8414 or permitted OIDC metadata ⟶ registration ⟶ authorization/token exchange; application callback handles user interaction.

invariants

Bind state, PKCE verifier, selected issuer, callback issuer, resource, redirect and scopes; reject replay/forged state; persist rotation before using refresh credentials; coalesce concurrent 401 flows.

failure

Discovery/issuer/registration/token/PKCE/callback/step-up failures remain explicit; repeated 401 retry is bounded; malicious discovery cannot relax credential or network policy.

- Legacy 2025-03-26 and 2025-06-18 are retained target ambitions. Deprecated dual-endpoint HTTP+SSE is optional compatibility; GET notification streams, Last-Event-ID, reconnect hints, ordered event replay and duplicate suppression require separate revision evidence. Parser carry-over alone does not prove exactly-once delivery.
- Logging retains an approved typed request-scoped emitter and legacy set-level behavior where defined. Arbitrary handler JSON logging is absent; no indirect sender is an approved emitter.
- Retained resource-read continuation, rich client interactions, form/URL elicitation outcomes, roots, sampling and extension methods are capability checked and revision scoped. Deprecated server-initiated behavior is not transplanted into modern wire methods.
- Tasks, Apps, DPoP, mTLS sender constraints, replay checks and workload identity require explicit pinned models and consumers. Public helpers retain paired transports, fake clock, scripted peers and configurable authorization metadata/JWKS server as unbuilt support.
- Relay is one distribution on one target; independent package or target splits require consumer and dependency evidence and change no capability responsibility. See **ADR-0001** and **ADR-0007**.

1.17 End-to-end walkthrough

A protected typed operation keeps protocol evidence separate from the business effect.

- An application owns Item and LookupError records and their input/output codecs. It defines lookup once, binds a handler with an explicit safe error renderer, creates server.new, and supplies its own principal/context policy.
- The protected HTTP endpoint validates its configuration before acquiring a listener. Its configured verifier checks the presented token and returns principal, audience and scopes; Relay admits exactly its resource and required scopes, then constructs native context in the serving request process.
- A client connects to the endpoint using one configuration and a view with a 10-second absolute deadline, cancellation token, correlation and optional order key. The request carries modern metadata and agreeing routing headers; schema and tool access refusal occur before the handler.
- The endpoint allocates an exchange, the runtime admits one invocation and the handler receives Call with native context, correlation and cancellation. It performs lookup or explicitly maps a durable invoke port, whose wait must fit inside the transport budget.
- A successful structured answer has native output plus content mirror; a safe application failure becomes ToolFailed; a required input round returns an originating continuation. On lost delivery the client receives MaybeSent and must consult business identity/reconciliation before retrying a mutation.
- A streaming peer disconnect closes publication and signals cancellation; the worker gets grace to stop work it owns elsewhere. Client close releases Relay-owned resources; endpoint stop disconnects its listener failure link before intentional termination. Shared HTTP Gun and business receipts retain their own application owners.
- One admitted-request deadline covers correlation, verification, context construction and finite execution; a callback that outlasts it loses dispatch authority. A caller needing a tighter bound sets the request timeout. No evidence above proves that an external business effect was undone.

Glossary

Every term the layer declares, with the context that owns it.

Admitted request deadline

owned by root

The single absolute deadline an HTTP endpoint starts at slot admission. It covers correlation, verification, context construction, finite execution and response writes. *Avoid*: request timeout — names the setter, not the interval it bounds.

Application context

owned by root

The caller-owned native value supplied to listing policy and handler invocation for one request.

Attestation

owned by root

A configured verifier’s assertion about a token’s principal, audiences, and scopes. Construction is not proof of cryptographic verification.

Bound tool

owned by root

A definition joined to an application handler before heterogeneous registration erases the handler’s native types.

Cancellation grace

owned by root

The bounded interval between cancellation of Relay-owned request or invocation work and forced termination of work that has not stopped.

Conformance evidence

owned by root

The assertions an external suite actually emitted for the scenarios it ran. It establishes only that tested scope.

Continuation

owned by root

The originating client, operation, arguments, signed request state, and required responses retained for a subsequent tool input round.

Correlation

owned by root

A Sinal value joining observations across a request and its downstream work. A carried value is untrusted telemetry rather than identity.

Cursor

owned by root

An opaque server-authenticated reference to a position in one listing family. Catalog and authorization-view binding remain required refinements.

Definition

owned by root

A reusable tool contract containing a validated name, an input codec, and either an output codec or content-only output.

Endpoint

owned by root

The HTTP owner of the current server description, admitted request slots, active streams, and an optional listener.

Exchange

owned by root

One inbound frame’s response destination and the lifecycle of output to that destination.

Grant

owned by root

An admitted principal, exact protected resource, and actual attested scopes accepted by Relay's resource-server policy.

Header acquisition

owned by root

The client's computation of dynamic request headers in a temporary request-owned process, bounded by the same budget as the request it serves.

Idempotency key

owned by root

An optional client assertion that repeated requests refer to one logical operation. The application determines identity using the authenticated principal and key.

Input contract

owned by root

A Blueprint contract loaded from a tool's application input schema for local argument validation. Admission to Blueprint's finite profile is separate from MCP protocol validity.

Invocation

owned by root

One admitted unit of application handler work, identified independently of its client's wire request id.

Protection

owned by root

The configured resource and endpoint scopes against which an attestation is admitted, together with authorization-server metadata.

Protocol schema

owned by root

The frozen revision's authority for JSON-RPC envelopes, method parameters, declarations, and result fields.

Protocol strategy

owned by root

A revision-bound codec and lifecycle pair, such as the modern per-request strategy or the retained initialized-session strategy, that shares native definitions and ports with other strategies.

Reducer

owned by root

A pure transition over one connection's protocol state that returns effects for an authoritative interpreter.

Request id

owned by root

The client-selected string or integer echoed inside a JSON-RPC response. It is neither execution identity nor authenticated authority.

Request owner

owned by root

The Relay coordinator holding dispatch, deadline, and cleanup authority for one admitted HTTP request. Its managed worker carries the callback process lifetime through the exchange. *Avoid*: callback worker — omits execution and cleanup ownership.

Runtime

owned by root

The actor owning one reducer history and supervising the invocation processes that interpret its effects.

Server description

owned by root

An immutable catalog of bound tools, resources, prompts, completion, identity, and tool access policy.

Sibling dependency

owned by root

A library of the same ecosystem that Relay imports. Published siblings use Hex version requirements, with resolved versions and checksums recorded in the manifest lock.

Submission evidence

owned by root

The client’s knowledge that a request was NotSent, MaybeSent, or completed its protocol exchange. It does not establish the application’s effect outcome.

Subscription

owned by root

One owned listen request with an acknowledged notification filter and a cancellable stream lifetime.

Tombstone

owned by root

A retained terminal identity suppressing late or repeated work during a bounded retention interval.

Verifier

owned by root

The application-supplied function that decides a bearer token’s validity and returns an attestation. Relay never parses tokens itself.